

A Breathing-Based Meditation Intervention for Patients With Major Depressive Disorder Following Inadequate Response to Antidepressants: A Randomized Controlled Pilot Study

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STRUCTURED ABSTRACT

Objective: To evaluate the feasibility, efficacy, and tolerability of an adjunctive breathing-based meditation program, Sudarshan Kriya Yoga (SKY), in patients with major depressive disorder (MDD) who have experienced an inadequate response to standard antidepressant therapy.

Methods: A two-arm, 8-week randomized controlled trial was conducted at the University of Pennsylvania. Adult outpatients with persistent MDD (HDRS-17 score ≥ 14 despite ≥ 8 weeks of stable antidepressant medication) were randomized to either adjunctive SKY ($n=13$) or a waitlist control group ($n=12$). The primary clinical endpoint was change in 17-item Hamilton Depression Rating Scale (HDRS-17) score from baseline to week 8. Secondary outcomes included Beck Depression Inventory (BDI) and Beck Anxiety Inventory (BAI).

Results: Twenty-five patients were randomized (intent-to-treat sample). Patients in the SKY intervention group demonstrated dramatic reductions in both depression and anxiety scores. Mean HDRS-17 score dropped by 9.77 points in SKY vs. an increase of 0.50 in control (difference = 10.27 points, $p = 0.0032$). BDI score decreased by 17.23 in SKY vs. 1.75 in controls ($p = 0.0101$). BAI anxiety score decreased by 5.44 in SKY vs. 0.25 in controls ($p = 0.0097$). Remission rate reached 36.4% in completers.

Conclusions: Sudarshan Kriya Yoga is a feasible, well-tolerated, and effective adjunctive intervention for patients with major depressive disorder who have had an inadequate response to antidepressants.

INTRODUCTION

Major depressive disorder (MDD) is a leading cause of disability worldwide. Despite advances in pharmacotherapy, more than 50% of patients with MDD fail to achieve full remission with first-line antidepressant medication, and approximately one-third remain chronically symptomatic despite multiple sequential medication trials. Consequently, there is an urgent need for safe, effective, and accessible non-pharmacological adjunctive treatments.

Sudarshan Kriya Yoga (SKY) is a standardized breathing-based meditative intervention involving specific rhythmic breathing cycles (slow, medium, and fast rhythms) followed by quiet awareness. Prior neurophysiological investigations suggest that rhythmic breath modulation engages the autonomic nervous system via vagal nerve stimulation, promoting parasympathetic tone, reducing cortisol and inflammatory markers, and enhancing prefrontal cortical activity.

METHODS & EXPERIMENTAL DESIGN

Study Participants: Outpatients aged 18-65 years with a primary diagnosis of nonpsychotic MDD according to DSM-IV criteria were recruited. Eligible participants had an HDRS-17 baseline score ≥ 14 and had received a stable therapeutic dose of an FDA-approved antidepressant for at least 8 weeks prior to enrollment without achieving clinical remission.

Intervention Protocol: Participants randomized to SKY completed a standardized 6-session experiential training program (taught by certified instructors) followed by weekly group follow-up sessions and daily home practice. The daily home practice consisted of approximately 30 minutes of rhythmic breathwork. Control participants remained on their

stable antidepressant pharmacotherapy on a waitlist status.

RESULTS & CLINICAL TRIAL OUTCOME DATA

Table 1: Change in Clinical Depression and Anxiety Scores From Baseline to Week 8 (RCT Intent-to-Treat Analysis)

Outcome Measure	Waitlist Control (n=12)	SKY Adjunctive (n=13)	Mean Difference	P-Value
17-Item Hamilton Depression (HDRS-17)	+0.50 (1.87)	-9.77 (1.90)	-10.27 pts	p = 0.0032
Beck Depression Inventory (BDI-II)	-1.75 (2.56)	-17.23 (2.70)	-15.48 pts	p = 0.0101
Beck Anxiety Inventory (BAI)	-0.25 (1.48)	-5.44 (1.60)	-5.19 pts	p = 0.0097
HDRS-17 Response Rate (>50% reduction)	8.33%	54.54% (Completers)	+46.21%	p = 0.0271
HDRS-17 Clinical Remission (Score <= 7)	8.33%	36.36% (Completers)	+28.03%	p = 0.1550

DISCUSSION & CLINICAL SAFETY

This randomized clinical trial provides empirical evidence that Sudarshan Kriya Yoga offers significant antidepressant and anxiolytic benefits when used as an adjunct to pharmacotherapy in patients with treatment-resistant MDD. The magnitude of symptom reduction observed on both clinician-rated (HDRS-17) and self-rated (BDI) instruments was substantial.

Importantly, the intervention demonstrated excellent safety and tolerability. There were no adverse psychiatric or somatic events reported during the trial, and completion rates among participants attending training were high. These findings highlight the potential of breath-based meditative practices to serve as accessible, non-pharmacological tools to enhance treatment outcomes in refractory mood disorders.